



American International University- Bangladesh (AIUB)

Faculty of Engineering (EEE)

Course Name:	Engineering Ethics	Course Code:	EEE 2216	Section: B
Semester:	Fall 2025-26	Submission date:	17 th Nov, 2025	
Item:	CO1: Evaluate the impact of professional engineering work in the solution of complex engineering problems/Ethical Dilemmas in societal and environmental contexts by giving judgments based on Code of Ethics.			

Your Serial #	ID	Student Name	Department	Marks
30	<u>22-49350-3</u>	<u>NAHIN, AENAN BIN ISLAM</u>	<u>CSE</u>	
31	<u>22-49385-3</u>	<u>NAYEEM, NAZMUL HASSAN</u>	<u>CSE</u>	
32	<u>22-49393-3</u>	<u>AMIN, EFAZ MASHRUR AMIN</u>	<u>CSE</u>	
33	<u>22-49419-3</u>	<u>MAHID, S M ABID HOSSAIN</u>	<u>CSE</u>	

Marking Rubrics (to be filled by Faculty)

Category	Proficient [10-9]	Good [8-6]	Average [5-3]	Poor [2-1]	Secured Marks
Explanation of Ethical Dilemma	Dilemma /problem to be considered critically is stated clearly and described comprehensively, delivering relevant information necessary for full understanding.	Dilemma /problem to be considered critically is stated, described, and clarified so that understanding is not seriously impeded by omissions.	Dilemma /problem to be considered critically is stated, but description leaves some terms undefined, ambiguities unexplored, boundaries undetermined,	Dilemma /problem to be considered critically is stated without clarification or description.	
Accepted practice areas in the engineering discipline (K7)	Identifies and relates all the solutions with the accepted practice in the engineering discipline.	Identifies and relates some of the solutions with the accepted practice in the engineering discipline.	Identifies and but does not relates the solutions with the accepted practice in the engineering discipline.	Fails to identify and does not relates the solutions with the accepted practice in the engineering discipline.	
Student's position (perspective, thesis/ hypothesis) (P1)	Specific position (perspective, hypothesis) is imaginative, considering the complexities of an issue. Limits of position (perspective, hypothesis) are acknowledged. Others' points of view and assumptions are synthesized within position (perspective, hypothesis).	Specific position (perspective, thesis/hypothesis) considers the complexities of an issue. Others' points of view and assumptions are acknowledged within position (perspective, hypothesis).	Specific position (perspective, hypothesis) acknowledges different sides of an issue.	Specific position (perspective, hypothesis) is stated, but is simplistic and obvious.	
Innovative Thinking or uniqueness (of idea, claim, question etc.) (P3)	Extends a novel or unique idea, question, format, or product to create new knowledge or knowledge that crosses boundaries.	Creates a novel or unique idea, question, format, or product.	Experiments with creating a novel or unique idea, question, format, or product.	Reformulates a collection of available ideas.	
Valid Conclusions and related outcomes including all factors (implications and consequences) (P7)	Conclusions and related outcomes (consequences and implications) are logical and reflect student's informed evaluation and ability to place evidence.	Conclusion is logically tied to a range of information, including opposing viewpoints; related outcomes (consequences and implications) are identified clearly.	Conclusion is logically tied to information (because information is chosen to fit the desired conclusion); some related outcomes (consequences and implications) are not clear.	Conclusion is inconsistently tied to some of the information discussed; related outcomes (consequences and implications) are oversimplified.	

For complete Similarity with other (Negative Marking will be imposed)

Comments:		Total Marks (Out of 50):	
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NVIDIA is a large company that sells computers, computer components, and software. Ralph is hired as an entry-level software engineer at NVIDIA. His first project was to assist in writing the code for NVIDIA's new hard disc controller. He had previously worked on a similar system interning at a start-up and had written a code which greatly enhanced the performance of their product. Ralph quietly re-uses this same code in the NVIDIA product and does not think to tell anyone that he has used the code from his last job. His manager is thrilled with the speed improvements this code brings to the product.

Before the product is released, it has to undergo a four-month long quality assurance process review. During the review of the product, it was found the code which Ralph developed had been copyrighted by the startup he had previously worked for. Even though Ralph had developed the code, his previous company still owned the intellectual property rights to it.

When his manager informed Ralph of the problem, Ralph admits he did not realize he had made a mistake because he was not familiar with copyright laws. Ralph then goes on to explain that the start-up he used to work for is now out of business and is unsure if NVIDIA would be able to get in contact with the owner of the copyright. If NVIDIA can't use Ralph's code, then it will have to rewrite the entire code of the product, delaying its release by many months.

Briefly discuss the above case and provide the options following the eight subsequent steps in resolving the Ethical Dilemma.

ANSWER:

Briefly discussion:

Ralph, a new engineer at NVIDIA, reused code he wrote at a previous start-up without telling anyone. Although he created it, the copyright legally belonged to his former company. During quality checks, NVIDIA discovered the code was copyrighted, meaning they cannot use it without permission. Because the start-up is now closed, getting permission is difficult, and NVIDIA may have to rewrite the code, causing major delays. The case shows the importance of understanding copyright laws and disclosing reused code.

Dilemma:

Should Ralph and NVIDIA keep quiet to avoid delaying the project, or should they be honest about the mistake and fix the code, even though it'll cost more time and money?"

Step-by-Step Ethical Resolution

1. Identify the Moral Dilemma

Conflict: Product release vs. Copyright law

Moral tension:

To avoid delays and issues, follow with ethical and legal guidelines or remain silent.

2. Gather All Relevant Facts

- Ralph wrote the code, but the old start-up owns the copyright.
- Although the business is closed, copyright is still in effect.
- The product release will be delayed if the code is rewritten.
- NVIDIA may face legal repercussions if it uses copyrighted code.

3. Identify Stakeholders

- Ralph
- NVIDIA (management, engineers, legal team)
- Copyright owner (old company)
- Customers
- Software engineering community

4. Refer to Professional Codes of Ethics

Examples (ACM/IEEE Code of Ethics):

- "Software engineers must respect creative property."
- "Software engineers must be trustworthy and honest."
- "Software engineers must stay safe."

Respect for copyright > Finishing the project on time

5. Consider the Options

Option A: Do nothing and just release the product.

Pros: No delay

Cons: It is illegal, it is unethical, and it may lead to a lawsuit.

Option B: Fix secretly later

Pros: Problem hidden

Cons: Dishonest, still illegal

Option C: Report the problem and try to obtain permission

Pros: Honest, legal

Cons: Possible delay

Option D: Delete and re-write the code

Pros: 100% ethical

Cons: Big delay, more cost

6. Evaluate the Best Course of Action

Using ethical reasoning:

- **Utilitarian:** Avoid harm --- Fix the problem legally.
- **Duty-based:** Must follow copyright law and professional code.

Best Choice: Option C --- Report the issue and follow legal process

If permission cannot be obtained ---- Proceed with Option D.

7. Take Action

- Ralph describes the exact use of the code.
- Notify project management and legal.
- Give up using the code that is safe by copyright.
- Try to get in touch with the owner of the copyright.
- If authorization is denied, rewrite the code.

8. Reflect and Learn

- Enhance instruction on intellectual property and copyright.
- Improve code review procedures to prevent such circumstances.
- Engineers should be encouraged to speak up early.
- Encourage software development teams to have an ethically conscious culture.